

Project Title: Adaptive Learning Multimodal AI Music Recommendation System

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Abstract

The "Adaptive Learning Multimodal AI Music Recommendation System" is a personalized music recommendation system that combines adaptive learning, feedback driven learning, and multi-modal input (psychological input and physiological input) to provide individualized recommendations to users. This system will make continuous updates to recommendations based on user preferences, body physiological signals (heart rate, stress level, etc.) at a specific point in time using a wearable smartwatch, as well as psychological inputs (DASS-21 (Depression Anxiety Stress Score-21), TIPI (Ten Item Personality Inventory), WHO-QOL (World Health Organisation-Quality Of Life)) through their questionnaire responses. To assist with personalizing the recommendations, users can select preferences such as current mood, genre and vibe for their music recommendations. Recommendations will be adapted based on the user's feedback. The AI engine uses a feedback loop based form of reinforcement learning to make adaptive updates and a hybrid model combining collaborative filtering with content based filtering to match the music characteristics of users with their known musical tastes and emotional moods. Hyperparameter optimization and systematic validation of the models result in optimal performance for both types of AI based track generation. AI generated music tracks can also be created to produce new personalized content for each user. Combining these multiple methods to create a framework for individualized recommendations is possible through adaptive learning techniques. Ongoing adaptation of the system based on changing circumstances provides ongoing improvement and engagement of users with the service; thus creating a strong, adaptable base for further development and future use of personalized media services through research.

SDG Alignment: This work contributes to Sustainable Development Goal (SDG) 3:

Good Health and Well-Being by promoting stress reduction, emotional regulation, and mental wellness through personalized music-based interventions.